

Barcode Checksum Rescue

Catch the fake barcode before the wrong part ships

At a glance: Catch the fake barcode before the warehouse ships the wrong part. Plan about 60 minutes for the core block; 4 to 6 teams of 4 to 5 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: error detection, check digits, data reliability, applied math (Understand).
- Apply the weighted mod-10 rule to verify and classify 12-digit UPC-A codes (Apply).
- Analyze weighted sums to detect corrupted codes without rejecting valid codes (Analyze).
- Defend each accept-or-reject verdict with complete UPC-A arithmetic (Evaluate).

MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- Printed barcode cards
- Dry erase boards
- Timers

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Stage the 12-digit UPC-A card deck, dry erase boards, markers, and timers at each team station.
- 02 Stage the worked UPC-A example, scoring sheet, and staff answer key; verify every corrupted card before use.
- 03 No special PPE is required. Keep cards and dry erase materials organized, and use only the verified 12-digit UPC-A deck.
- 04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Catch the fake barcode before the warehouse ships the wrong part." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Teams work one valid example and explain the odd-position x 3 rule before the timed deck begins.

Phase 3 Constraints

0:12 - 0:18

Use 12-digit UPC-A only: odd positions x 3, even positions x 1, and accept only totals that are multiples of 10.

Phase 4 Verify UPC-A codes

0:18 - 0:38

Teams verify and classify 12-digit UPC-A codes. Circulate to check position numbering and complete arithmetic without revealing verdicts.

Phase 5 Test and record

Teams record accept-or-reject verdicts and flag suspected corruptions; staff score correct detections and false alarms from the key.

Phase 6 Verification audit

Teams double-check each flagged card, show one full weighted sum, and explain why a single changed digit is detected.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with arithmetic evidence, erase boards, and return the card deck in order.

TROUBLESHOOTING

Problem: A team weights digits from the wrong end. **Fix:** Number the 12 UPC-A positions from the left and shade odd positions before recalculating.

Problem: A total does not match the key. **Fix:** Recheck every digit, the odd-position x 3 subtotal, and the final mod-10 test.

Problem: A valid code is flagged. **Fix:** Require a second complete calculation before counting the rejection as final.

Problem: A team finishes early. **Fix:** Have them create one single-digit corruption and prove that the rule catches it.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to check digits catch errors.

Tier 2 (calculating but not reasoning):

- "Which positions are multiplied by 3, and how can you keep that pattern straight?"

Tier 3 (ready to extend):

- "Create one single-digit error, recalculate the weighted total, and explain why the code now fails."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- How does one extra digit catch a typo?
- Why do false alarms matter in a warehouse?
- Where do you see check digits in daily life?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Correct detections	45
Low false alarms	20
Rule explanation	20
Teamwork	15
Total	100

CLEANUP AND SOURCES

Return barcode cards in order, erase boards, cap markers, return timers, and collect score slips.

SOURCES Computer science unplugged error-detection concepts